

AutoParts Hub

An e-commerce platform for car parts and repair services — built for car owners, enthusiasts, and repair shops.

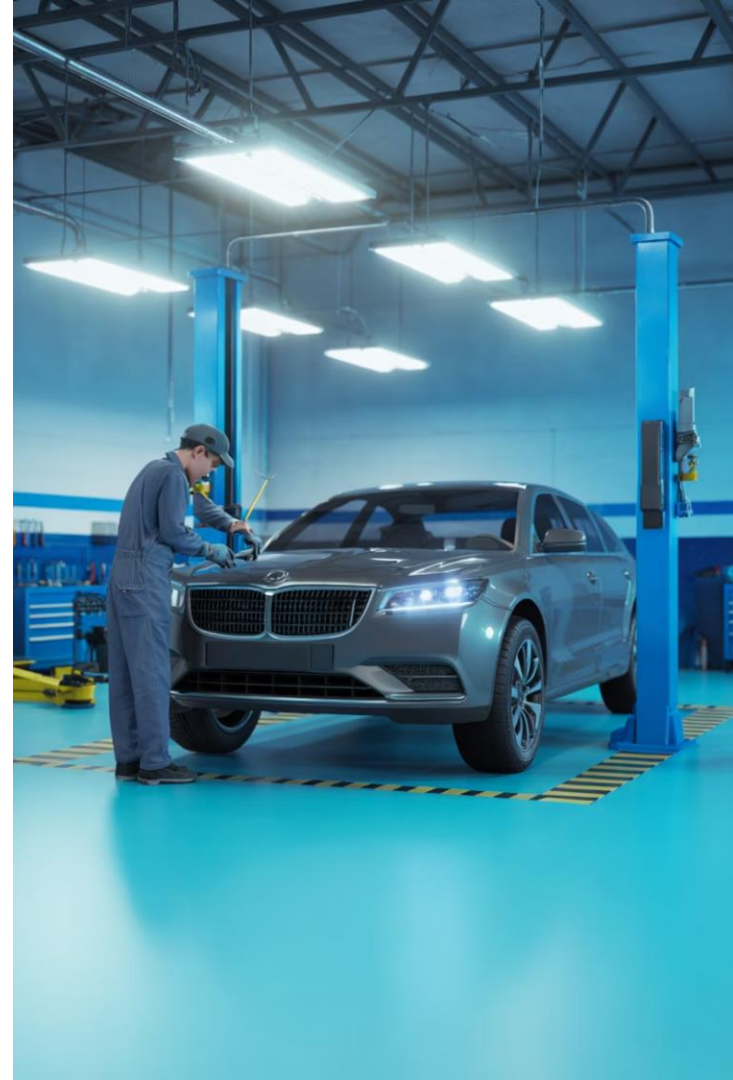
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- Course: Mobile Commerce
- Dr. Mohammed Haitham

Arab Academy for Science, Technology & Maritime Transport

Project files: Auto Parts Hub Complete Project (zip file)

<https://drive.google.com/file/d/1xa0E9DcsdNFOv8ICUTGOxIAq3rKTqQN2/view?usp=sharing>



What Is AutoParts Hub?

AutoParts Hub is a comprehensive e-commerce platform for car parts and repair services, designed to simplify vehicle maintenance for every car owner.



Browse & Purchase

Shop car parts from trusted vendors with a seamless checkout experience.



Mechanic on Demand

Request professional mechanics directly to your location.



AI Assistant

AI-powered car advice to help you make informed decisions.



Favorites & Tracking

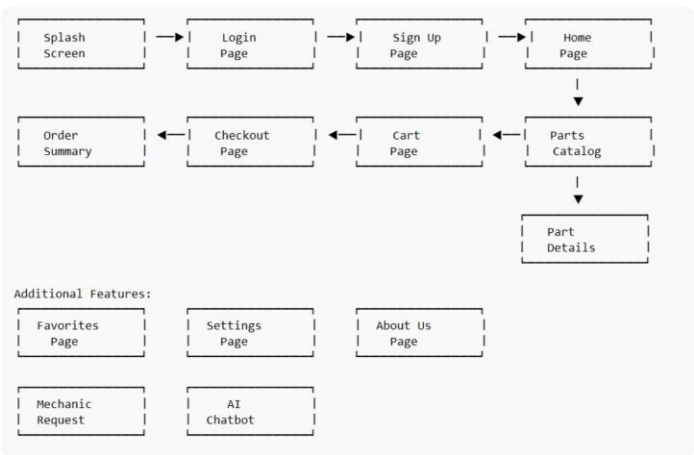
Save favorite products and track orders in real-time.



Target Audience: Car owners, car enthusiasts, and repair shops.

Flow Diagram & UI/UX Animations

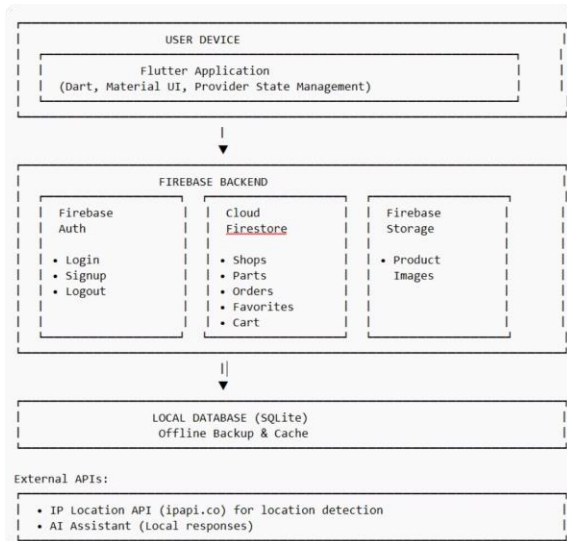
The diagram below illustrates the complete user journey through the AutoParts Hub application, from onboarding to order completion and service requests.



## UI/UX Animations##		
Animation Type	Location	Effect
Page Transition	All navigation	Fade/Slide
Heart Button	Part Details	Scale 1.3x
Hero Image	Product images	Cross-fade
Shimmer Loading	Parts Catalog	Skeleton glow
Button Press	All buttons	Scale down
Category Chip	Parts Catalog	Color change

System Architecture

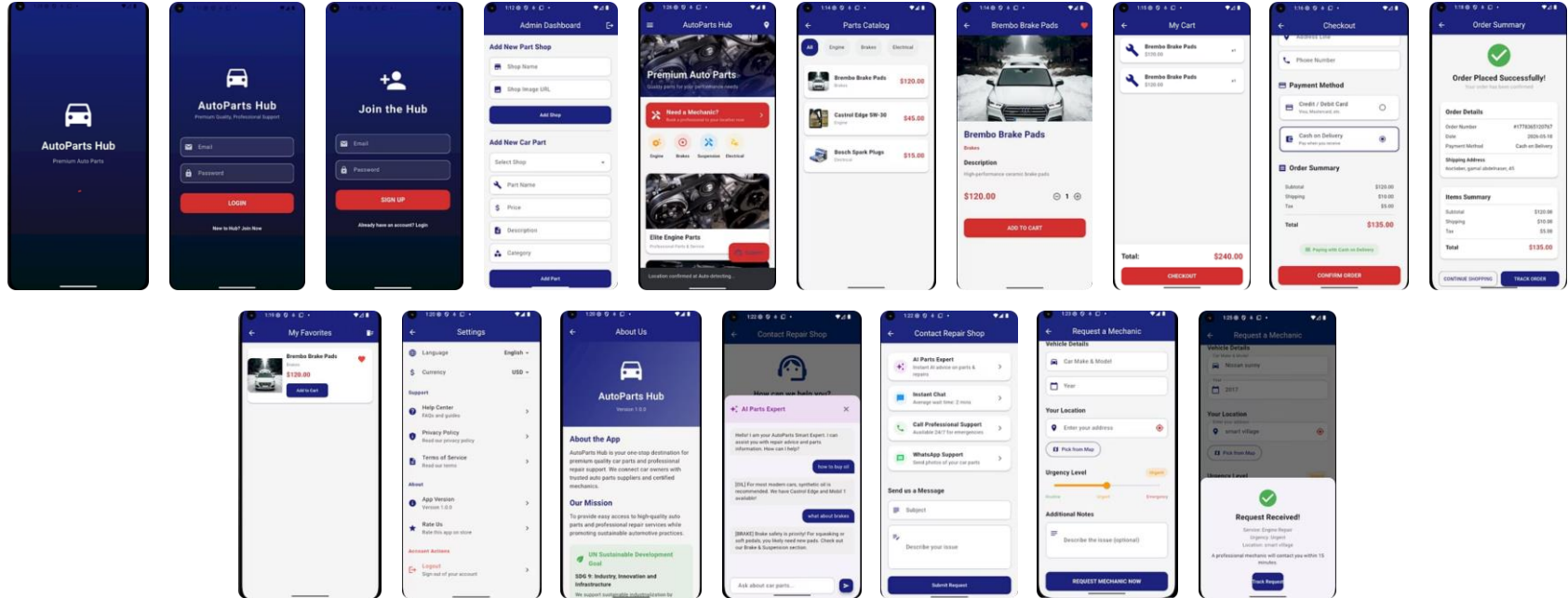
The system architecture diagram below outlines the technical structure of AutoParts Hub, showing how frontend, backend, and data layers interact.



🏠 > favorites > wfn7kcSvG1xdx...		
🔊 (default)	📁 favorites	📄 wfn7kcSvG1xdx8MfvSLw
+ Start collection	+ Add document	+ Start collection
⋮ favorites >	wfn7kcSvG1xdx8MfvSLw >	+ Add field
parts		addedAt: 9 May 2026 at 03:02:38 UTC+3
shops		category: "Brakes"
users		imageUrl: "https://images.unsplash.com/photo-1517524008697-84bbe3c3fd98?w=500&q=80"
		partId: "T4Ykqqmoc4Eaj6O5SAzL"
		partName: "Brembo Brake Pads"
		price: 120
		userId: "OEctDq6xT5RxpdhUGuOxgeCgDliI"

Screenshots

A visual walkthrough of the AutoParts Hub mobile application interface, showcasing the home screen, product browsing, AI assistant, mechanic booking, order tracking, and more.



UN SDG Alignment

Our Commitment

AutoParts Hub is designed with sustainability at its core, contributing to multiple United Nations Sustainable Development Goals through eco-conscious features and community-driven services.

"Our app makes car maintenance more sustainable by connecting users with local suppliers and promoting eco-friendly parts."

How We Contribute

- Promotes sustainable industrialization
- Supports eco-friendly auto parts
- Reduces waste through quality products
- Connects users with local repair shops
- Reduces transportation emissions
- Encourages innovation in auto parts industry

Future Work

AutoParts Hub follows a phased release strategy to progressively enhance functionality and user experience.

1

Short-Term

Next Release

- Push notifications for order updates
- Real-time order tracking
- Multi-language support (Arabic/English)
- Dark mode theme

2

Medium-Term

Second Release

- Payment gateway integration (Stripe, PayPal, Fawry)
- User reviews and ratings for products
- Loyalty points system
- Social media login (Google, Apple)

3

Long-Term

Third Release

- AR part viewer (see parts in 3D)
- Voice search for products
- Mechanic scheduling calendar
- Admin analytics dashboard
- Chat between users and mechanics
- AI-powered part recommendations

Technologies Used



Frontend

- Flutter SDK 3.41.4
- Dart Programming Language
- Material Design 3



Backend

- Firebase Authentication
- Cloud Firestore (NoSQL Database)
- Firebase Storage



Local Database

- SQLite (sqflite package)



APIs & Services

- IP Location API (ipapi.co)
- HTTP Client for API calls



Development Tools

- VS Code
- Android Studio
- Git & GitHub
- Firebase Console



State Management

- setState (UI state)
- FutureBuilder (Async data)

Team Members & Roles

AutoParts Hub was developed by a collaborative team, each contributing specialized expertise to deliver a complete mobile commerce solution.



Project Lead & Backend Developer

Firebase, Firestore integration and backend architecture



UI/UX Designer

App design, theme, and user experience



Database Developer

SQLite and local storage implementation



AI Integration

Chatbot and AI-powered features



QA Tester

Testing and bug fixes

Conclusion

AutoParts Hub successfully delivers a complete, production-ready mobile commerce experience for the automotive market.



E-Commerce

Complete e-commerce functionality



Firebase Backend

Firebase cloud backend



AI Assistant

AI-powered assistant



Mechanic Integration

Professional mechanic integration



User-Friendly UI

User-friendly interface



UN SDG Aligned

UN SDG alignment

Thank You!

Questions? We love to hear from you. (GitHub: <https://github.com/abdurahman-mohamed-hassan/AutoParts-Hub.git>)